AD7031 Optimization Under Uncertainty

# Homework 2

**Due Oct 31, start of class**

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Write down your solutions and submit them as a single PDF to `lossof0706@gmail.com`.
**Your solutions must be handwritten.**

## Problem 1. Mean and covariance of a random vector **(25 points)**

We have three independent identically distributed (i.i.d.) return data points:

| | Tech | Energy | Bond |
|---|---|---|---|
| $\xi^1$ | 0.00 | 0.04 | 0.03 |
| $\xi^2$ | 0.06 | 0.10 | 0.03 |
| $\xi^3$ | 0.12 | 0.07 | 0.03 |

Note that we use $\widehat{\mathbb{E}}[\cdot]$, $\widehat{\mathrm{Var}}[\cdot]$ and $\widehat{\mathrm{Cov}}[\cdot]$ for empirical estimates, to distinguish them from their true counterparts $\mathbb{E}[\cdot]$, $\mathrm{Var}[\cdot]$ and $\mathrm{Cov}[\cdot]$.

(a) **(8 pts) By hand**, compute the empirical mean $\widehat{\mu} = \widehat{\mathbb{E}}[\tilde{\xi}]$ and the empirical covariance matrix $\widehat{\Sigma} = \widehat{\mathrm{Cov}}[\tilde{\xi}]$, and state the dimensions of $\widehat{\mu}$ and $\widehat{\Sigma}$.

(b) **(5 pts)** We used only three data points. In general, with $N$ i.i.d. samples of $\tilde{\xi}$, the empirical mean and covariance matrix can be written $\widehat{\mu}_N$ and $\widehat{\Sigma}_N$. Consider the claim

$$\lim_{N\to\infty} \widehat{\mu}_N = \mathbb{E}[\tilde{\xi}] = \mu, \qquad \lim_{N\to\infty} \widehat{\Sigma}_N = \mathrm{Cov}[\tilde{\xi}] = \Sigma.$$

What does it mean? Explain in plain Korean. Is it true? If so, why?

(c) **(7 pts)** Suppose your optimal portfolio is $x^\star = (x_1^\star, x_2^\star, x_3^\star)^\top$. From the table, compute, in terms of $x^\star$,

- the empirical mean $\widehat{\mathbb{E}}[\tilde{\xi}^\top x^\star]$;
- the empirical variance $\widehat{\mathrm{Var}}[\tilde{\xi}^\top x^\star]$.

(d) **(5 pts)** Show that

- $\widehat{\mathbb{E}}[\tilde{\xi}^\top x^\star] = \widehat{\mu}^\top x^\star$;
- $\widehat{\mathrm{Var}}[\tilde{\xi}^\top x^\star] = x^{\star\top} \widehat{\Sigma}\, x^\star$.

*hint: we discussed this in class.*

## Problem 2. Linear program duality **(25 points)**

Consider the following primal linear program and its dual, in the canonical form of the lecture note:

$$\text{(P)} \quad \begin{aligned} \min_{x\in\mathbb{R}^n} \quad & c^\top x \\ \text{s.t.} \quad & Ax \le b \end{aligned} \qquad\qquad \text{(D)} \quad \begin{aligned} \max_{\theta\in\mathbb{R}^m} \quad & -b^\top \theta \\ \text{s.t.} \quad & c + A^\top \theta = 0 \\ & \theta \ge 0 \end{aligned}$$

(a) **(8 pts)** Derive the dual problem (D) from the primal problem (P). **Show all the steps clearly.**

(b) **(7 pts)** Consider the primal problem

$$\begin{aligned} \min_{x\in\mathbb{R}^2} \quad & -4x_1 - 5x_2 \\ \text{s.t.} \quad & 3x_1 + 2x_2 \le 60 \\ & x_1 + 3x_2 \le 48 \\ & x_1 \ge 3 \\ & x_1,\ x_2 \ge 0. \end{aligned}$$

- Give $c$, $A$, $b$.
- Write the dual problem (D) numerically.

*hint: follow the same steps as in (a), this time with these numbers.*

(c) **(7 pts)** Derive the primal problem (P) from the dual problem (D). **Show all the steps clearly.**

*hint: nothing really changes from the derivation in (a).*

(d) **(3 pts)** Let $x^\star$ and $\theta^\star$ be optimal solutions of the primal problem (P) and the dual problem (D), respectively. What can you say about $c^\top x^\star$ and $b^\top \theta^\star$?

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## Problem 3. Conic program duality **(25 points)**

Consider

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & x^\top \Sigma x \\ \text{s.t.} \quad & \mu^\top x \geq r \\ & e^\top x \leq 1 \\ & x \geq 0 \end{aligned}$$

where $e = [1, 1, \ldots, 1]^\top$ is the all-ones vector and $r$ is the target return.

(a) **(12 pts)** Recall the second-order cone

$$\mathrm{SOC}_Q^{n+1} = \big\{(x,t) \in \mathbb{R}^n \times \mathbb{R} \;:\; x^\top Q\, x \leq t^2,\; t \geq 0\big\}, \qquad Q \succ 0.$$

Write the model in the conic form $\min c^\top z$ s.t. $Az \leq b$, $z \in C$.

- Give $z$ (the decision variable), $c$, $A$, $b$.
- Write $C$.

*hint 1: first, use the epigraph reformulation with a new decision variable $t$, then define $z = [x,\ t]^\top$.*

*hint 2: minimizing $x^\top \Sigma x$ and minimizing $\sqrt{x^\top \Sigma x}$ give the same $x$.*

(b) **(13 pts)** With $\Sigma = I$, the dual problem of the second-order cone program in (a) is

$$\begin{aligned} \max \quad & r\,\theta_1 - \theta_2 \\ \text{s.t.} \quad & \theta_1,\ \theta_2 \geq 0, \quad \theta_3 \in \mathbb{R}^n_+ \\ & \|\theta_2 e - \theta_1 \mu - \theta_3\|_2 \ \leq\ 1. \end{aligned}$$

Derive it. **Show all the steps clearly.**